

12 Vector-Valued Functions



12.3 Velocity and Acceleration

Objectives

- Describe the velocity and acceleration associated with a vector-valued function.
- Use a vector-valued function to analyze projectile motion.

Velocity and Acceleration

As an object moves along a curve in the plane, the coordinates x and y of its center of mass are each functions of time t .

Rather than using the letters f and g to represent these two functions, it is convenient to write $x = x(t)$ and $y = y(t)$.

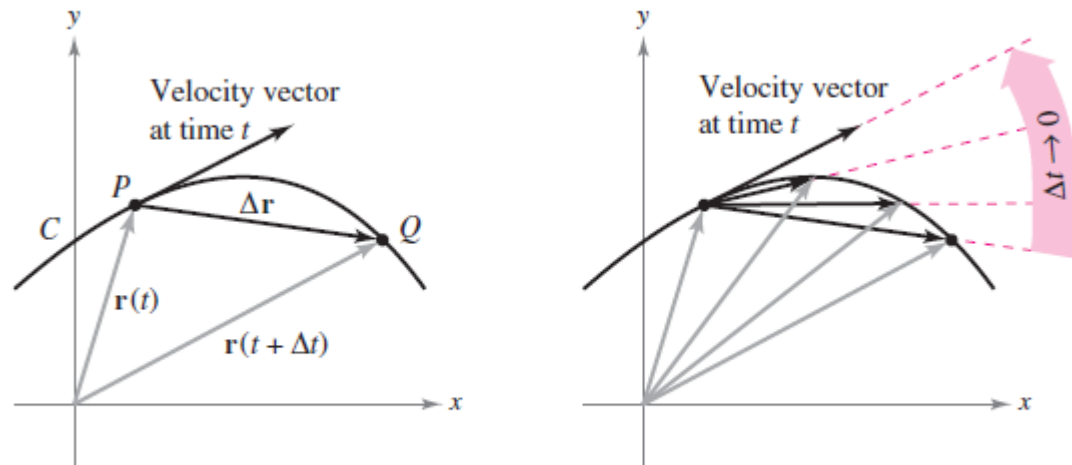
So, the position vector $\mathbf{r}(t)$ takes the form

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}. \quad \text{Position vector}$$

Velocity and Acceleration

To find the velocity and acceleration vectors at a given time t , consider a point $Q(x(t + \Delta t), y(t + \Delta t))$ that is approaching the point $P(x(t), y(t))$ along the curve C given by

$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$, as shown in Figure 12.11.



As $\Delta t \rightarrow 0$, $\frac{\Delta \mathbf{r}}{\Delta t}$ approaches the velocity vector.

Figure 12.11

Velocity and Acceleration

As $\Delta t \rightarrow 0$, the direction of the vector $\overrightarrow{PQ}$ (denoted by $\Delta \mathbf{r}$) approaches the *direction of motion* at time t .

$$\Delta \mathbf{r} = \mathbf{r}(t + \Delta t) - \mathbf{r}(t)$$

$$\frac{\Delta \mathbf{r}}{\Delta t} = \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{r}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

When this limit exists, it is defined as the **velocity vector** or **tangent vector** to the curve at point P .

Velocity and Acceleration

Note that this is the same limit used to define $\mathbf{r}'(t)$. So, the direction of $\mathbf{r}'(t)$ gives the direction of motion at time t .

Moreover, the magnitude of the vector $\mathbf{r}'(t)$

$$\|\mathbf{r}'(t)\| = \|x'(t)\mathbf{i} + y'(t)\mathbf{j}\| = \sqrt{[x'(t)]^2 + [y'(t)]^2}$$

gives the **speed** of the object at time t .

Velocity and Acceleration

Similar to how $\mathbf{r}'(t)$ is used to find velocity, you can use $\mathbf{r}''(t)$ to find acceleration, as indicated in the following definitions.

Definitions of Velocity and Acceleration

If x and y are twice-differentiable functions of t , and $\mathbf{r}$ is a vector-valued function given by $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$, then the velocity vector, acceleration vector, and speed at time t are as follows.

$$\text{Velocity} = \mathbf{v}(t) = \mathbf{r}'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j}$$

$$\text{Acceleration} = \mathbf{a}(t) = \mathbf{r}''(t) = x''(t)\mathbf{i} + y''(t)\mathbf{j}$$

$$\text{Speed} = \|\mathbf{v}(t)\| = \|\mathbf{r}'(t)\| = \sqrt{[x'(t)]^2 + [y'(t)]^2}$$

Velocity and Acceleration

For motion along a space curve, the definitions are similar.

That is, for

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$$

you have the following.

$$\textbf{Velocity} = \mathbf{v}(t) = \mathbf{r}'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j} + z'(t)\mathbf{k}$$

$$\textbf{Acceleration} = \mathbf{a}(t) = \mathbf{r}''(t) = x''(t)\mathbf{i} + y''(t)\mathbf{j} + z''(t)\mathbf{k}$$

$$\textbf{Speed} = \|\mathbf{v}(t)\| = \|\mathbf{r}'(t)\| = \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}.$$

Example 1 – Velocity and Acceleration Along a Plane Curve

Find the (a) velocity vector, (b) speed, and (c) acceleration vector for the particle that moves along the plane curve C described by

$$\mathbf{r}(t) = 2 \sin \frac{t}{2} \mathbf{i} + 2 \cos \frac{t}{2} \mathbf{j}. \quad \text{Position vector}$$

Solution:

$$\text{a. } \mathbf{v}(t) = \mathbf{r}'(t) = \cos \frac{t}{2} \mathbf{i} - \sin \frac{t}{2} \mathbf{j} \quad \text{Velocity vector}$$

$$\text{b. } \|\mathbf{r}'(t)\| = \sqrt{\cos^2 \frac{t}{2} + \sin^2 \frac{t}{2}} = 1 \quad \text{Speed (at any time)}$$

$$\text{c. } \mathbf{a}(t) = \mathbf{r}''(t) = -\frac{1}{2} \sin \frac{t}{2} \mathbf{i} - \frac{1}{2} \cos \frac{t}{2} \mathbf{j} \quad \text{Acceleration vector}$$



Projectile Motion

Projectile Motion

You now have the machinery to derive the parametric equations for the path of a projectile.

Assume that gravity is the only force acting on the projectile after it is launched. So, the motion occurs in a vertical plane, which can be represented by the xy -coordinate system with the origin as a point on Earth's surface, as shown in Figure 12.17.

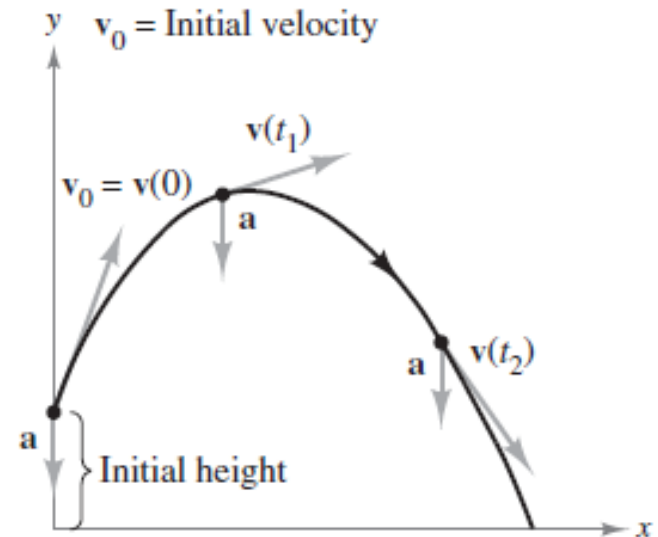


Figure 12.17

Projectile Motion

For a projectile of mass m , the force due to gravity is

$$\mathbf{F} = -mg\mathbf{j} \quad \text{Force due to gravity}$$

where the acceleration due to gravity is

$g = 32$ feet per second per second, or
9.8 meters per second per second.

By **Newton's Second Law of Motion**, this same force produces an acceleration $\mathbf{a} = \mathbf{a}(t)$, and satisfies the equation $\mathbf{F} = m\mathbf{a}$.

Consequently, the acceleration of the projectile is given by $m\mathbf{a} = -mg\mathbf{j}$, which implies that

$$\mathbf{a} = -g\mathbf{j}. \quad \text{Acceleration of projectile}$$

Example 5 – *Derivation of the Position Vector for a Projectile*

A projectile of mass m is launched from an initial position $\mathbf{r}_0$ with an initial velocity $\mathbf{v}_0$. Find its position vector as a function of time.

Solution:

Begin with the acceleration $\mathbf{a}(t) = -g\mathbf{j}$ and integrate twice.

$$\mathbf{v}(t) = \int \mathbf{a}(t) dt = \int -g\mathbf{j} dt = -gt\mathbf{j} + \mathbf{C}_1$$

$$\mathbf{r}(t) = \int \mathbf{v}(t) dt = \int (-gt\mathbf{j} + \mathbf{C}_1) dt = -\frac{1}{2} gt^2\mathbf{j} + \mathbf{C}_1 t + \mathbf{C}_2$$

Example 5 – *Solution*

cont'd

You can use the initial conditions $\mathbf{v}(0) = \mathbf{v}_0$ and $\mathbf{r}(0) = \mathbf{r}_0$ to solve for the constant vectors $\mathbf{C}_1$ and $\mathbf{C}_2$.

Doing this produces $\mathbf{C}_1 = \mathbf{v}_0$ and $\mathbf{C}_2 = \mathbf{r}_0$.

Therefore, the position vector is

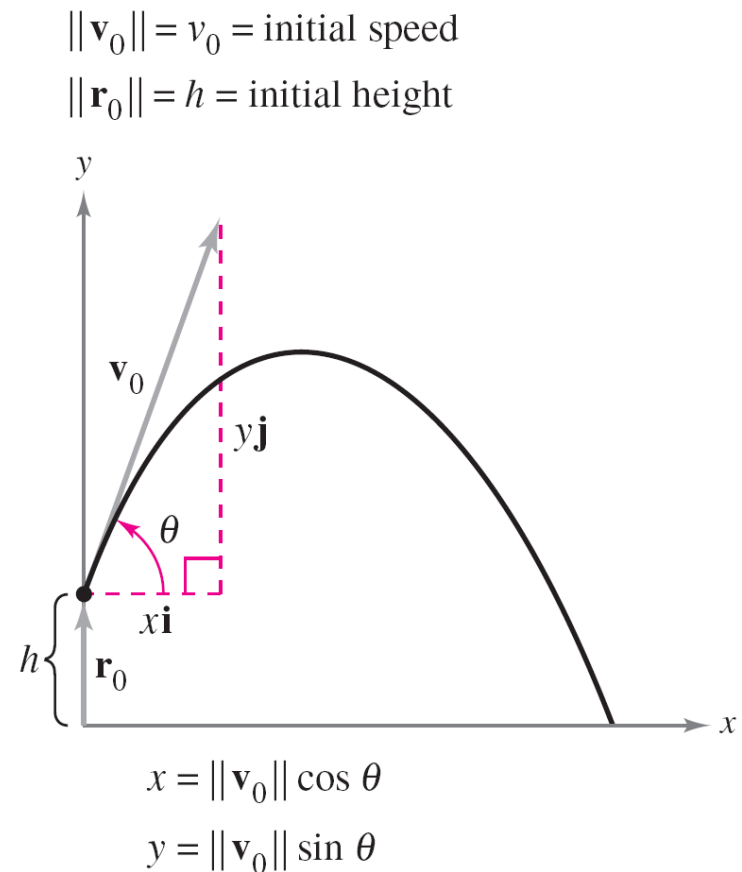
$$\mathbf{r}(t) = -\frac{1}{2}gt^2\mathbf{j} + t\mathbf{v}_0 + \mathbf{r}_0.$$

Position vector

Projectile Motion

In many projectile problems, the constant vectors $\mathbf{r}_0$ and $\mathbf{v}_0$ are not given explicitly.

Often you are given the initial height h , the initial speed v_0 , and the angle θ at which the projectile is launched, as shown in Figure 12.18.



Projectile Motion

From the given height, you can deduce that $\mathbf{r}_0 = h\mathbf{j}$.
Because the speed gives the magnitude of the initial velocity, it follows that $v_0 = \|\mathbf{v}_0\|$ and you can write

$$\begin{aligned}\mathbf{v}_0 &= x\mathbf{i} + y\mathbf{j} \\ &= (\|\mathbf{v}_0\| \cos \theta)\mathbf{i} + (\|\mathbf{v}_0\| \sin \theta)\mathbf{j} \\ &= v_0 \cos \theta \mathbf{i} + v_0 \sin \theta \mathbf{j}.\end{aligned}$$

Projectile Motion

So, the position vector can be written in the form

$$\mathbf{r}(t) = -\frac{1}{2}gt^2\mathbf{j} + t\mathbf{v}_0 + \mathbf{r}_0$$

Position vector

$$= -\frac{1}{2}gt^2\mathbf{j} + tv_0 \cos \theta \mathbf{i} + tv_0 \sin \theta \mathbf{j} + h\mathbf{j}$$

$$= (v_0 \cos \theta)t\mathbf{i} + \left[h + (v_0 \sin \theta)t - \frac{1}{2}gt^2 \right]\mathbf{j}.$$

Projectile Motion

THEOREM 12.3 Position Vector for a Projectile

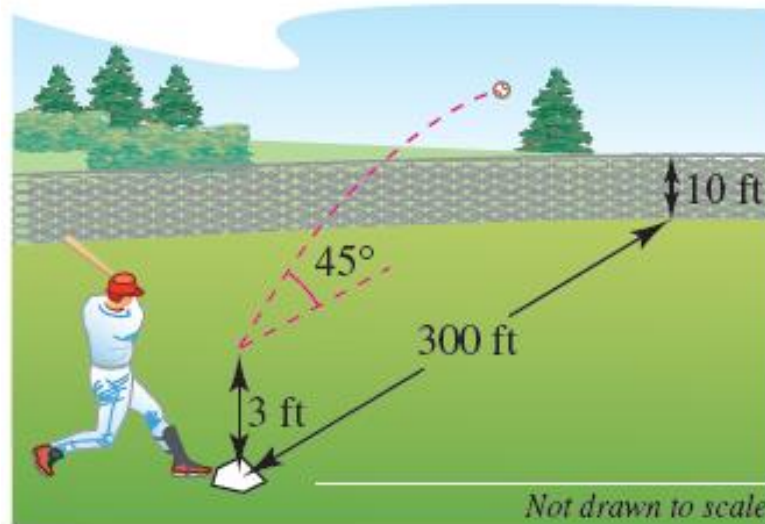
Neglecting air resistance, the path of a projectile launched from an initial height h with initial speed v_0 and angle of elevation θ is described by the vector function

$$\mathbf{r}(t) = (v_0 \cos \theta)t\mathbf{i} + \left[h + (v_0 \sin \theta)t - \frac{1}{2}gt^2\right]\mathbf{j}$$

where g is the acceleration due to gravity.

Example 6 – *Describing the Path of a Baseball*

A baseball is hit 3 feet above ground level at 100 feet per second and at an angle of 45° with respect to the ground, as shown in Figure 12.19. Find the maximum height reached by the baseball. Will it clear a 10-foot-high fence located 300 feet from home plate?



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Figure 12.19

Example 6 – *Solution*

You are given $h = 3$, $v_0 = 100$, and $\theta = 45^\circ$.

So, using Theorem 12.3 with $g = 32$ feet per second per second produces

$$\begin{aligned}\mathbf{r}(t) &= \left(100 \cos \frac{\pi}{4}\right)t \mathbf{i} + \left[3 + \left(100 \sin \frac{\pi}{4}\right)t - 16t^2\right] \mathbf{j} \\ &= (50\sqrt{2}t) \mathbf{i} + (3 + 50\sqrt{2}t - 16t^2) \mathbf{j}\end{aligned}$$

The velocity vector is

$$\mathbf{v}(t) = \mathbf{r}'(t) = 50\sqrt{2} \mathbf{i} + (50\sqrt{2} - 32t) \mathbf{j}.$$

Example 6 – *Solution*

cont'd

The maximum height occurs when

$$y'(t) = 50\sqrt{2} - 32t$$

is equal to 0, which implies that

$$t = \frac{25\sqrt{2}}{16} \approx 2.21 \text{ seconds.}$$

So, the maximum height reached by the ball is

$$\begin{aligned} y &= 3 + 50\sqrt{2}\left(\frac{25\sqrt{2}}{16}\right) - 16\left(\frac{25\sqrt{2}}{16}\right)^2 \\ &= \frac{649}{8} \end{aligned}$$

≈ 81 feet.

Maximum height when $t \approx 2.21$ seconds

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Example 6 – *Solution*

cont'd

The ball is 300 feet from where it was hit when

$$x(t) = 300 \quad \Rightarrow \quad 50\sqrt{2}t = 300.$$

Solving this equation for t produces $t = 3\sqrt{2} \approx 4.24$ seconds.

At this time, the height of the ball is

$$y = 3 + 50\sqrt{2}(3\sqrt{2}) - 16(3\sqrt{2})^2$$

$$= 303 - 288$$

$$= 15 \text{ feet.} \quad \text{Height when } t \approx 4.24 \text{ seconds}$$

Therefore, the ball clears the 10-foot fence for a home run.